# Shivam Kumar Verma

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## Experience

**poiro.com** **Jul 2026 – Present**
*Fullstack Developer Intern* *Python, FastAPI, MongoDB, Redis, Celery, Next.js, TypeScript, Cloudflare R2*

- **AI Workflow Orchestration:** Built the node and group system behind the AI workflow builder: each workflow is a DAG of generation steps with typed ports and a defined execution order, and groups bundle steps into nested sub-workflows that run as one unit.
- **Fail-Closed Spend Caps:** Integrated Pixabay, Unsplash, and Pinterest behind one adapter contract with per-org Redis caching, and capped provider calls fail-closed at **500 per org per day** and **5,000 per day globally** so no workflow can run up vendor costs.
- **Digital Asset Service:** Designed the org-scoped MongoDB models and FastAPI backend for one asset service shared by **3 apps**, migrated existing assets into it, and added upload-time duplicate blocking with pHash and dHash image similarity.

**klimb.io** **Nov 2025 – May 2026**
*Full Stack Developer Intern* *TypeScript, Express.js, PostgreSQL, Redis, AWS*

- **Auditable Usage Ledger:** Replaced on/off feature flags with a PostgreSQL credit ledger that records every use of **6+ AI features**, which made usage-based billing possible.

## Projects

**CodeSM | GitHub | Live | *Node.js, BullMQ, Redis, PostgreSQL, Docker***

- **Sandboxed Code Execution:** Architected a code platform that runs untrusted submissions in **6 languages** in Docker with no network, a read-only filesystem, and CPU, memory, and time limits, graded on BullMQ workers off the request path.
- **AI Interview Practice:** Added Gemini-generated hints (**4** per problem) and resume-personalized interview questions with written-answer scoring, voiced through Murf AI.

**DeepDoc | GitHub | *RAG, TypeScript, Azure OpenAI, Pinecone, PostgreSQL***

- **LLM Evaluation Harness:** Created a **36-case** evaluation harness (**6 multi-turn**, including Hinglish and prompt-injection cases) that replays retrieval sweeps, A/B tests prompts, and separates indexing, retrieval, and generation failures; it caught a bug that blanked every retrieved chunk.

**Hireflow | GitHub | *Node.js, BullMQ, PostgreSQL, Gemini, GPT-4o-mini***

- **Guardrailed LLM Pipeline:** Shipped an AI job-application engine that parses listings into structured stack and seniority with Gemini, rewrites resume bullets with GPT-4o-mini, and rejects any rewritten skill not in the real profile; used for real applications.

## Technical Skills

**Languages:** Python, TypeScript, SQL, C++
**AI Engineering:** LLM evaluation, RAG, context engineering, embeddings and vector search, query rewriting, prompt A/B testing
**Backend:** FastAPI, Node.js, Express.js, REST APIs, Celery, BullMQ, WebSockets
**Data & Infrastructure:** Data modelling, PostgreSQL, MongoDB, Redis, Pinecone, Docker, Kubernetes, AWS (EC2, S3)
**Frontend & Tooling:** React, Next.js, Playwright, Git, Linux, CI pipelines

## Education

**Indian Institute of Information Technology Nagpur** **Nov 2022 – May 2026**
*B.Tech in Computer Science and Engineering* *Nagpur, Maharashtra*

## Achievements

- **Competitive Programming: Top 5% globally** on LeetCode (**1800+ rating**, 800+ problems solved) · **3-star** on CodeChef (1700+) · Pupil on Codeforces.
- **Hackathon: 3rd place** at Tantrafiesta 2024, IIIT Nagpur.